
BYTES ARE NOT LATENCY: ALLOCATED SIZE, MEASURED TRAFFIC, DECODE SPEED, AND QUALITY OF KV-CACHE QUANTIZATION ON A SINGLE BLACKWELL GPU

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A REPRODUCIBILITY

The results were produced at repository commit [anonymized]. The study used model `meta-llama/Llama-3.1-8B-Instruct` at revision `0e9e39f249a16976918f6564b8830bc894c89659`. Performance runs used measurement container [anonymized], and quality runs used its derivative [anonymized]. Performance data are identified by the tag [anonymized]. The hardware manifest, per-run configuration, raw samples, and five independent process measurements per point are retained under content-addressed roots with indefinite retention: full-scan wall-clock root [anonymized], profiler root [anonymized], modeling root [anonymized], and joint-results root [anonymized]. The KIVI and KVQuant source patches are stored in the repository with their manifests. Two CPU-only packages run without a GPU or network access: the modeling package (Python standard library only) regenerates the modeling audit, figures, and predictor examples, and the joint-results package (Python with pyarrow) regenerates the per-stage outcome table, the joined quality–performance table, and the empty table of qualified configurations byte-for-byte.

Modeling. Beyond the limits in Section 4, fold-level coefficient vectors were not saved, so held-out predictions are tied to the stored out-of-fold table and the fitting code.

Run accounting. The full scan comprises 2,670 planned runs (534 points $\times$ 5 processes): 2,205 completed runs and 465 predicted capacity-infeasible. Thirty-eight runs were replaced after infrastructure failures, each linked to the run it replaces, and no valid, slow, or unstable observation was rerun; output checksums, kernel paths, and allocations agree across processes. In the eager-versus-Graph experiment, 10 of 120 conditions were predicted infeasible and 105 were stable; five eager conditions (three BF16, two TQ-3bit) exceeded the 3% CV threshold and were kept as

unstable, excluded from fitted comparisons, and not rerun. The quality–performance join matches runs by exact configuration identity and retains all 2,670 planned runs, with no unmatched or duplicated identities.

Artifact availability. The code, configurations, and method patches are released at [anonymized URL] as release tag v1.0 (default branch) under the Apache-2.0 license; vendored and patched third-party code keeps its upstream terms, listed in the repository’s NOTICE file. The protocol, decision records, and full research history are on the main branch at commit [anonymized]. The two CPU reproduction packages are released as GitHub Release assets. The raw measurements are available from the author on request. Model weights are gated under the Llama 3.1 Community License and are not redistributed.

Modeling package (Python standard library only), run from the package root:

```
python3 reproduce.py reproduce --package . \
--output <output_dir>
```

Joint quality–performance package (Python with pyarrow), run from the package root:

```
python3 reproduce.py --reproduce . \
--output <output_dir>
```

In both cases `<output_dir>` must not exist yet.

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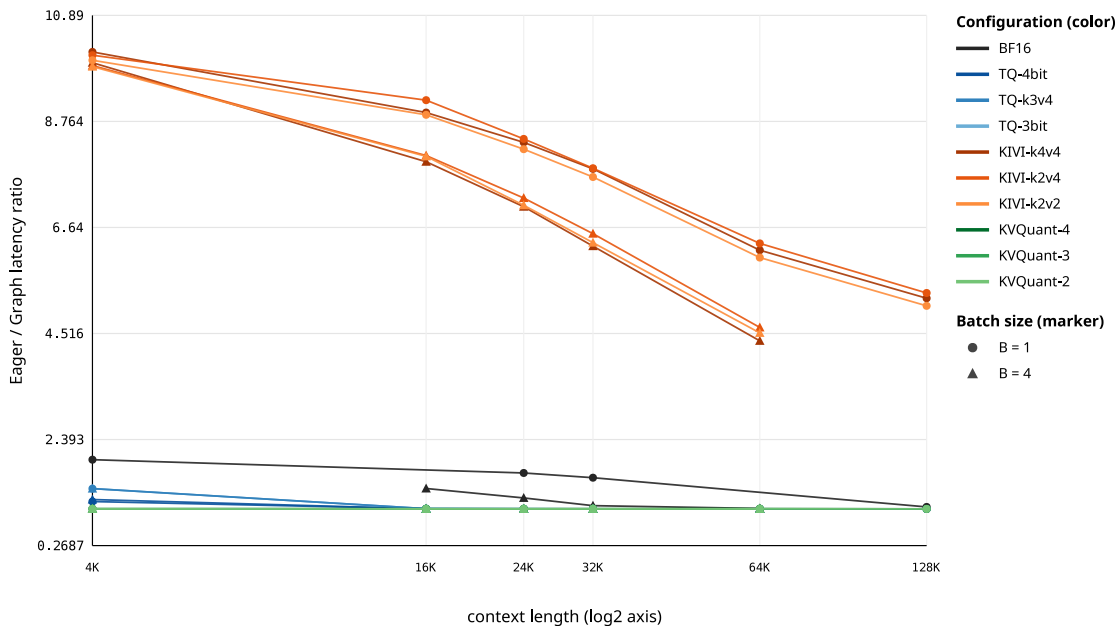


Figure A1. Eager-to-Graph latency ratio versus context for each configuration at $B = 1$ and $B = 4$ (within-configuration mechanism experiment). KIVI’s ratio starts near 10 and declines with context. BF16 starts near 2. TurboQuant and KVQuant stay close to 1.

B PROTOCOL TIMELINE

Date (2026)	Event
Jul 22	Scope, model, and performance measurement protocol fixed; post-performance quality protocol (stages, gates, margins) preregistered
Jul 25–31	Measurement container fixed; per-method and unified validation gates passed
Aug 22–26	Pilot scan; knee-densification design (84 adaptive contexts) frozen and executed; an earlier blocked pilot attempt contributed no timing
Aug 26–30	Eager/Graph mechanism experiment; profiler subset
Aug 31–Sep 16	Full scan executed; wall-clock primary analysis completed; an earlier stopped scan attempt contributed no timing
Sep 16–17	Modeling and CPU reproduction package
Sep 17–18	Performance data frozen; exact quality inputs and mechanical choices fixed
Sep 19	Correctness stage: all ten configurations fail cross-batch invariance; BF16 diagnosis
Sep 19	Physical- $B = 1$ amendment approved, before any perplexity score
Sep 19–21	Fast perplexity
Sep 21–27	Full perplexity (KIVI-k4v4)
Sep 27–Oct 2	LongBench-E (KIVI-k4v4)
Oct 2–4	LongBench v2 no-CoT (KIVI-k4v4)
Oct 4	Joint validation and scoped quality–performance join

preregistered wall-clock statistics from the same raw files without rerunning any timing, and all timing numbers in this paper use wall-clock time.

The first full-scan summary computed ratios from CUDA-event medians. A later append-only analysis re-derived the